

Navigators Technical Documentation

Navigators — Offline Architecture & Synchronization Protocols

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1. Fundamental Conceptual Clarification: Internet Outage vs. GNSS Outage

A common misconception in mobile engineering is conflating cellular internet connectivity with satellite positioning. In Navigators, these are strictly independent physical and operational states:

	GNSS Available	GNSS Denied (Outage)
Internet Online	State A: Standard Run Full GNSS + Live Server Telemetry Streaming	State B: Tunnel Drive AI Dead Reckoning Cellular Connected
Internet Offline	State C: Remote Highway GNSS Satellite Fix Zero Cellular/Server	State D: Total Denied Pure Autonomous Offline AI DR + Local Vector Map

- **Internet Outage (Offline Mode):** Cellular data or Wi-Fi is unavailable (Airplane Mode, rural dead-zones, disaster scenarios). The application must boot from local disk cache, run inference on-device, and persist user interactions locally.
- **GNSS Outage (Satellite Denial):** Satellite signals are blocked or jammed (tunnels, underground parkings, dense high-rise urban canyons). The application must compute vehicle velocity and position using internal sensors and dead reckoning.
- **Total Denial (State D):** Both internet and GNSS are absent. **Navigators is specifically architected to maintain continuous navigation under State D.**

2. Complete Offline Runtime Architecture

flowchart TD

```

subgraph BrowserEnvironment["Client Browser / Mobile PWA Runtime"]
    SW["Service Worker (CacheStorage)"]
    HTML["index.html / index.css / Leaflet"]
    WASM["ONNX Runtime WebAssembly (`ort.wasm`)"]
    Model["TCN Model Weights (`model.onnx`)"]
    MapData["Vector Road Network (`road_network.json`)"]
    IndexedDB["LocalStorage / IndexedDB Sync Queue"]
    Engine["Dead Reckoning & EKF Engine"]
end

```

end

```

subgraph HardwareSensors["Smartphone Hardware"]
    IMU["Accelerometer & Gyroscope"]
end

```

end

```

SW -->|Serve Static Shell| HTML
SW -->|Serve WASM Binary| WASM
SW -->|Serve Neural Graph| Model
SW -->|Serve Vector Map| MapData
IMU -->|High-Rate Kinematics| Engine
WASM -->|SIMD Execution| Engine
Model -->|Trained Weights| WASM
MapData -->|Centerline Graphs| Engine
Engine -->|Offline User Edits / POIs| IndexedDB

```

2.1 Service Worker Cache Strategy (`simulator/sw.js`)

The Service Worker implements a **Cache-First** strategy for all core execution assets: 1.

`simulator/index.html`, `index.css`, `app.js` 2. `simulator/vendor/onnxruntime/ort.wasm.min.js`, `ort-wasm-simd-threaded.wasm` 3. `simulator/vendor/leaflet/leaflet.js`, `leaflet.css` 4. `simulator/model.onnx` 5. `simulator/data/road_network.json`

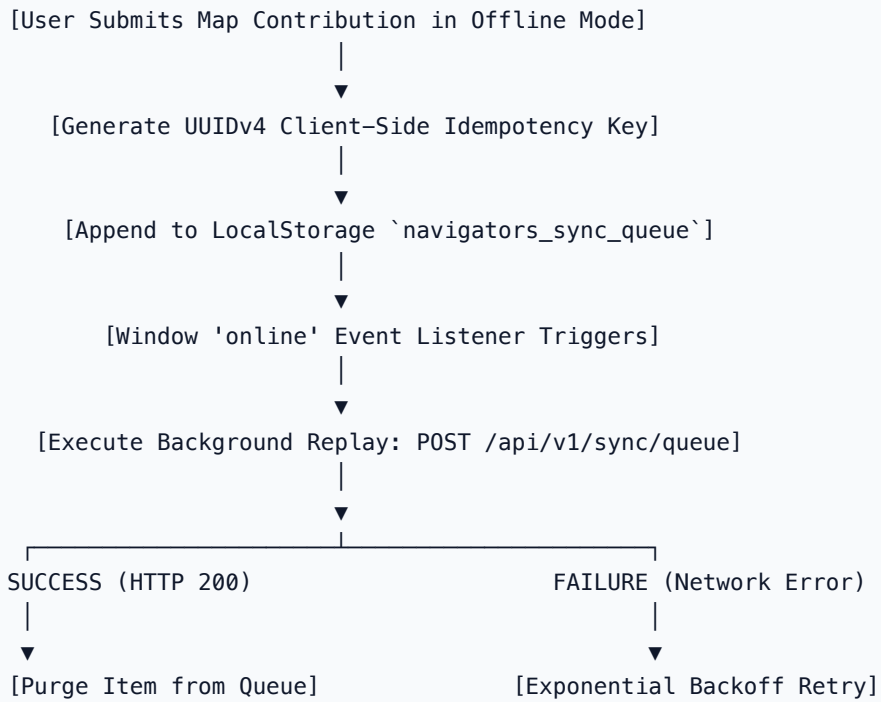
When an update is published, the service worker verifies checksums, swaps the cache atomically in the background, and prompts for a seamless reload.

3. Local Data Persistence & Vector Storage

1. **Map Data:** Road centerline polylines, intersection nodes, and speed attributes are serialized as a lightweight GeoJSON vector payload (1.1 MB) stored directly in client storage. The application does not require external tile server requests to display the route corridor.
2. **Local Places Cache:** Community-approved places of interest (fuel stations, hospitals, charging points) are stored locally in `LocalMap` (`simulator/local_map.js`), enabling instant search and turn-by-turn routing without backend queries.

4. Offline Contribution Queue & Sync Protocol

When a user submits a map update, POI creation, or road incident while operating in offline mode:



4.1 Idempotency Key Enforcement

Every queued submission is assigned a unique `client_sync_id` (UUIDv4) upon creation. When the client reconnects and replays the sync queue to `POST /api/v1/sync/queue`: - The backend checks `sync_queue` in SQLite. - If `client_sync_id` already exists, the server returns the existing result without creating duplicate records or modifying state twice.

4.2 Conflict Resolution

- **Canonical Map Edits:** If an offline contribution edits a place that was concurrently modified or deleted on the server, the server preserves the server record and routes the offline edit into the Moderator review queue with a conflict flag (`status = 'pending_review'`).
- **Telemetry Ingestion:** Ingested sensor sessions are appended immutably to the dataset repository; conflicts do not occur because raw trips are uniquely keyed per device and timestamp.

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